Capstone Project

Price & Power: A 64-Year Analysis of Egypt's Economic Resilience and Purchasing Power (1960–2024)

GUIDEBOOK

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Project Timeline Summary

Task Name	Start Date	Days Spent (Duration)
Problem Definition	04-03-2026	10
Data Preparation	14-03-2026	10
Data Modeling	24-03-2026	10
Exploratory Analysis	04-04-2016	10
Visualization	24-04-2026	20
Insights & Recommendations	04-05-2026	10
Presentation Q&A	19-05-2026	15

The project follows a structured three-month timeline beginning in early March and extending through mid-June 2026. The initial phase focuses on defining the problem and conducting research, followed immediately by data collection, cleaning, and modeling throughout March. Upon establishing a validated data model, exploratory analysis is conducted in early April to uncover initial patterns. The visualization phase is a primary focus from late April through mid-May, during which the dashboard is designed, implemented, and refined over a twenty-day period. Subsequently, insights and recommendations are developed and validated with the team in early June. The final stage of the project is dedicated to preparing the presentation, rehearsing, and handling the final Q&A session to deliver the project by mid-June.

This initiative is offered by the Ministry of Communications and Information Technology under the name *Digital Egypt Pioneers Initiative (DEPI)*, in collaboration with private training partners such as CLS Learning Solutions, Leading 365, Berlitz, and Elharefa. Launched in 2023, the program aims to equip students for the freelance and digital job market, providing specialized tracks in technical skills, soft skills, freelancing, and English.

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Introduction

Over the past six decades, Egypt's economy has experienced major transformations driven by monetary policies, exchange rate fluctuations, and external economic shocks. Although income levels have increased over time, this growth does not necessarily reflect an improvement in citizens' real living standards.

This project, **"Cost of Living"**, analyzes Egypt's economic performance from **1960 to 2024**, focusing on inflation and its relationship with key indicators such as income, unemployment, interest rates, exchange rate, trade balance, and foreign reserves.

The main objective is to address the **"illusion of nominal values"**, where rising income creates a false perception of improved welfare, while inflation erodes purchasing power. By measuring real purchasing power and related indicators, the project provides a more accurate view of economic well-being.

Using an interactive dashboard, the project demonstrates how weaknesses in economic fundamentals—such as trade deficits or currency depreciation—lead to higher inflation, reduced purchasing power, and increased economic pressure on citizens.

Ultimately, the project connects macroeconomic performance with everyday financial reality, offering insights that support better understanding of past trends, policy impacts, and future economic challenges.

Chapter 1 Overview: Data Collection and Data Preparation

1.1 Data Collection

The dataset used in this project was collected from World Bank Open Data, specifically from the World Development Indicators (WDI) database. The data covers Egypt's economic indicators from 1960 to 2024 and was selected to analyze the relationship between macroeconomic performance and citizens' purchasing power.

The selected indicators include:

- Inflation, consumer prices (annual %)
- Official exchange rate (LCU per US$, period average)
- Exports of goods and services (current US$)
- Imports of goods and services (current US$)
- GDP per capita (current US$)
- Lending interest rate (%)
- Broad money (% of GDP)
- Total reserves including gold (current US$)
- Unemployment, total (% of total labor force) modeled ILO estimate

These indicators were chosen because they represent the main economic factors affecting cost of living, currency stability, purchasing power, and overall economic pressure on citizens.

1.2 Data Preparation Using Power Query

After collecting the raw data, the dataset was imported into Power BI and transformed using Power Query. Since World Bank data is usually exported in a wide format, where years appear as separate columns, several transformation steps were required to make the data suitable for analysis.

Note: The following transformation steps were applied consistently to all economic indicators (Inflation, Exchange Rate, Exports, Imports, GDP per Capita, Interest Rate, Money Supply, Reserves, and Unemployment).
The only exception is the scaling step, which was applied only to percentage-based indicators.

step	Power Query Code	Description	Output
1	Table.PromoteHeaders(Source, [PromoteAllScalars=true])	Promotes the first row of the imported World Bank file to become column headers.	Initial column names are created from the first row.
2	Table.TransformColumnTypes(#"Promoted Headers", {...})	Changes the data types of the raw columns: text columns for metadata and numeric columns for yearly values.	Columns become readable by Power Query with correct initial data types.
3	Table.Skip(#"Changed Type", 3)	Removes the first three unnecessary metadata rows from the World Bank export file.	Cleaner dataset starting from the actual data table.

4	Table.PromoteHeaders(#"Removed Top Rows", [PromoteAllScalars=true])	Promotes the correct row again after removing metadata rows.	Proper headers appear, such as Country Name, Country Code, Indicator Name, Indicator Code, and years.
5	Table.TransformColumnTypes(#"Promoted Headers1", {...})	Converts country and indicator fields to text, and year columns from 1960 to 2025 to numeric values.	Dataset is ready for calculations and timeseries analysis.
6	Table.UnpivotOtherColumns(#"Changed Type1", {"Country Name", "Country Code", "Indicator Name", "Indicator Code"}, "Attribute", "Value")	Converts the dataset from wide format into long format by turning year columns into rows.	Two new columns are created: Attribute and Value.

7	Table.RenameColumns(#"Unpivoted Other Columns", {{"Attribute", "Years"}, {"Value", "Value"}})	Renames the generated columns to meaningful names.	Standardized naming across all datasets.
8	Table.TransformColumnTypes(#"Renamed Columns", {{"Years", Int64.Type}, {"Value", type number}})	Converts Years to integer and values to numeric.	Data becomes ready for sorting and analysis.
9	Table.SelectRows(#"Changed Type2", each ([Country Name] = "Egypt, Arab Rep."))	Filters the dataset to include only Egypt's records.	The dataset becomes focused only on Egypt.
10	Table.SelectRows(#"Filtered Rows", each [Years] <> null and [Years] <> "")	Removes rows where the year value is missing or invalid.	Clean timeseries data without blank years.
11	Table.RenameColumns(...)	Renames the indicator column to a clearer final name.	Final column name becomes Inflation Rate.

Special Step (Applied Only to Percentage-Based Indicators)

STEP	POWER QUERY CODE	DESCRIPTION	OUTPUT

13	Table.TransformColumns(#"Renamed Columns1", {{"Indicator", each _ / 100, type number}})	Converts percentage values into decimal format. Applied only to **Inflation Rate, Unemployment Rate, and Lending Interest Rate**.	Example: 10% → 0.10, ensuring correct calculations and formatting in Power BI.

Literature Review

1. Overview

This project is grounded in established economic theories that explain the relationship between macroeconomic indicators and citizens' living standards. The analysis focuses on how inflation, exchange rates, income, and labor market conditions interact to influence real purchasing power.

The literature review highlights the key theoretical frameworks that support the logic and structure of the analysis.

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2. Monetarist Theory

Monetarist theory, primarily associated with **Milton Friedman**, suggests that inflation is mainly caused by an increase in the money supply.

According to this theory:

"Inflation is always and everywhere a monetary phenomenon."

In the context of this project, this theory explains the relationship between **broad money (M2)** and inflation levels. An increase in money supply without a corresponding increase in production leads to rising prices.

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3. Purchasing Power Theory

Purchasing Power theory emphasizes that the real value of income depends on the price level of goods and services.

This means that:

- An increase in nominal income does not necessarily indicate improved living standards
- The actual well-being of individuals depends on their ability to purchase goods and services

This concept is central to the project, as it focuses on measuring the **real impact of inflation on citizens** rather than relying on nominal values.

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4. Phillips Curve

The Phillips Curve describes the relationship between inflation and unemployment.

Traditionally, it suggests that:

- Lower unemployment may lead to higher inflation
- Higher unemployment may reduce inflation

In this project, the relationship between inflation and unemployment is analyzed to understand economic pressure on citizens, especially during periods of economic instability.

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5. Exchange Rate Pass-Through Theory

This theory explains how changes in the exchange rate affect domestic prices.

When the local currency depreciates:

- The cost of imported goods increases
- Inflation rises due to higher production and consumption costs

This relationship is particularly relevant for economies that rely on imports, where exchange rate fluctuations have a direct impact on inflation.

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6. Misery Index

The Misery Index is a simple but effective indicator that combines:

- Inflation rate
- Unemployment rate

It is used to measure the overall economic pressure on citizens.

Higher values of the index indicate:

- Increased cost of living
- Reduced job opportunities
- Higher economic stress

This concept supports the project's focus on linking macroeconomic conditions with real-life citizen impact.

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7. Economic Stability and Trade Theory

Economic stability is strongly influenced by a country's external sector, including:

- Exports
- Imports
- Foreign reserves

A country with:

- Strong exports
- Sufficient reserves is better able to maintain currency stability and control inflation.
- Conversely:
- Trade deficits and low reserves increase vulnerability to economic shocks

This theory explains the connection between external economic performance and internal economic conditions.

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8. Research Gap

Most existing studies analyze economic indicators separately, focusing on individual metrics such as inflation, income, or unemployment.

However, this project addresses a key gap by:

- Integrating multiple indicators into a unified analytical model
- Linking macroeconomic performance directly to citizens' purchasing power
- Providing a time-series analysis over a long historical period (1960–2024)

Chapter 2: Database Design & Data Modeling

2.1 Overview

In this project, the data model was designed to support the analysis of Egypt's macroeconomic indicators over time, covering the period from **1960 to 2024**.

Since the dataset consists of multiple independent economic indicators (such as inflation, unemployment, exchange rate, exports, imports, reserves, and interest rates), it was necessary to identify a **common dimension** to unify the model.

The only shared attribute across all tables is **Years**, which represents the time dimension. Therefore, a central table was created to serve as the foundation of the data model.

2.2 Master Calendar Table

To standardize time-based analysis, a central dimension table named **DIM Master_Calendar** was created using DAX:

STEP	POWER QUERY CODE
1	DIM Master_Calendar = GENERATESERIES(1960, 2024, 1)

This table contains a continuous list of years from 1960 to 2024 and serves as the **central time dimension** in the model.

Why the Master Calendar Was Created

Without a unified calendar table:

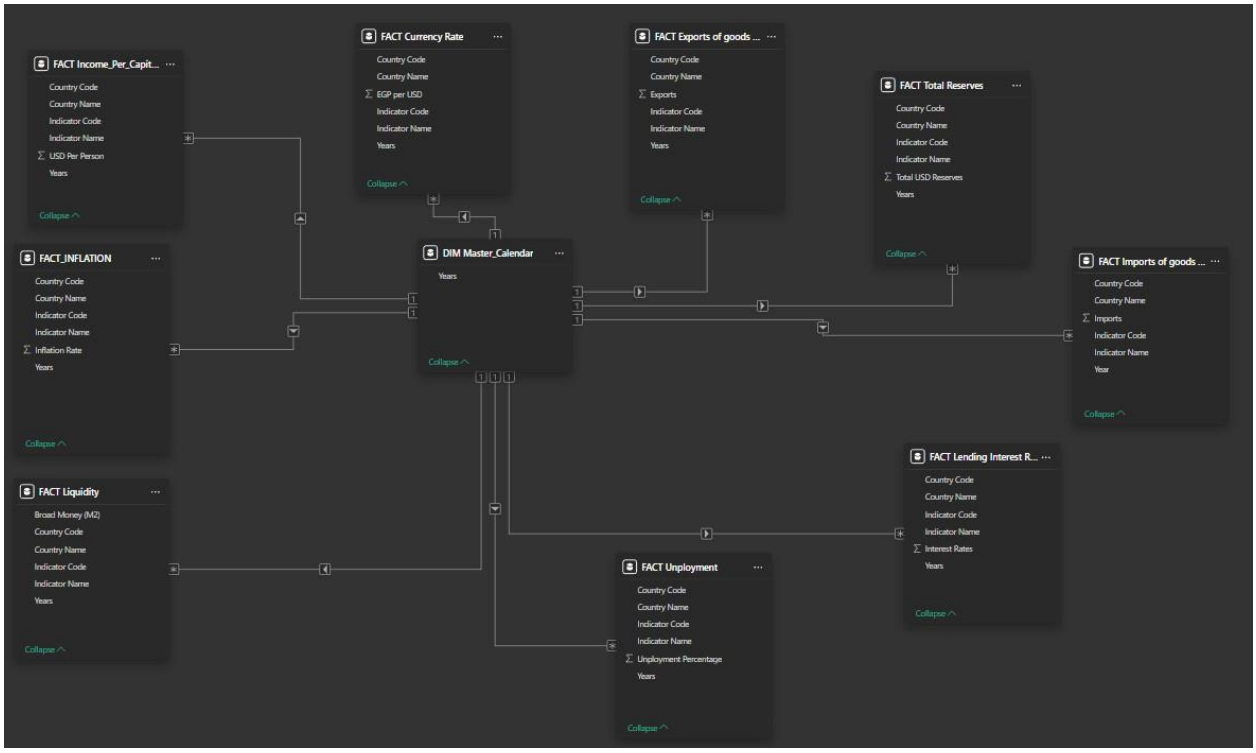
- Each dataset would operate independently
- Cross-indicator comparisons would be difficult
- Time-based filtering would not be consistent
- Calculations across multiple tables could produce incorrect results

By introducing the Master Calendar:

- A single time reference controls all datasets
- All indicators can be analyzed within the same time frame
- Filtering by year applies consistently across all tables
- The model becomes structured, scalable, and easier to maintain

Schema Structure :

The data model follows a Star Schema design.



DIM Master_Calendar is the central dimension table

All other tables are **Fact Tables** connected through the Years column

2.3 UI/UX & Dashboard Design

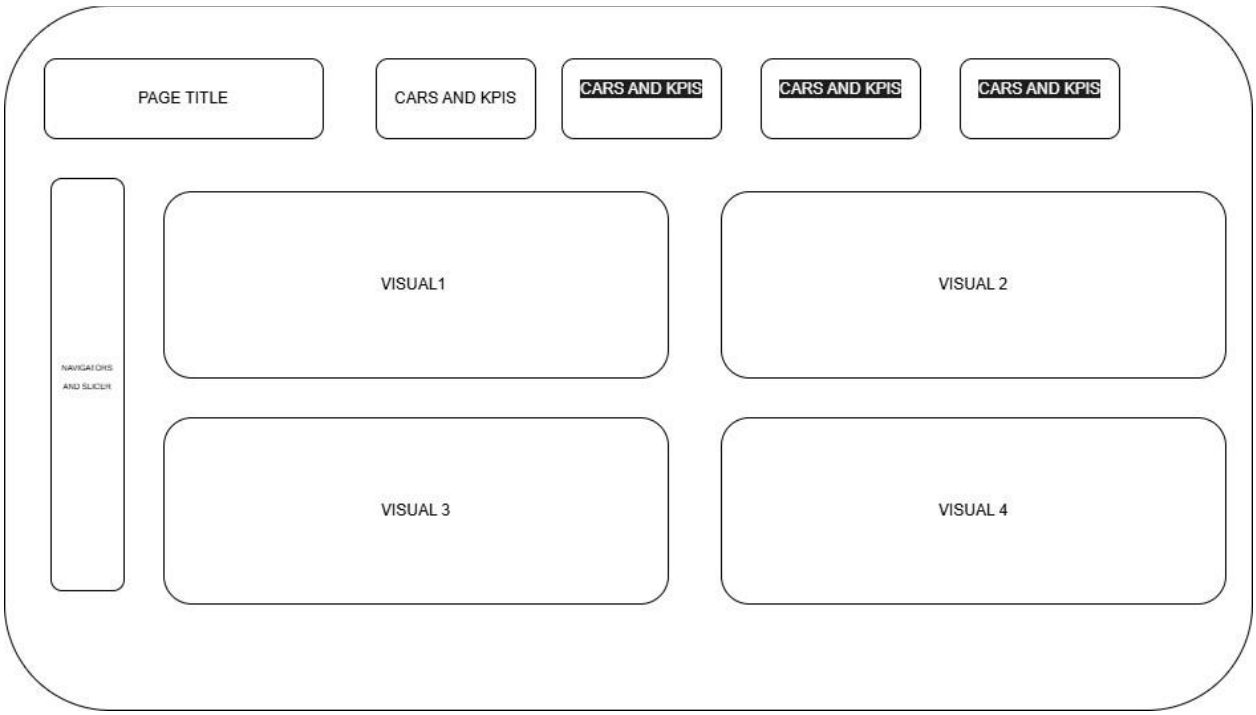
Design Overview

The dashboard was designed to provide a **clear, intuitive, and data-driven user experience**, enabling users to easily explore Egypt's economic indicators and understand their impact on purchasing power.

The design focuses on:

- Simplicity
- Clarity
- Consistency
- Analytical storytelling

The goal is to allow users (decision-makers, analysts, or general users) to quickly interpret complex economic data without confusion.



Layout Structure

The dashboard follows a structured layout divided into four main sections:

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1. Header Section

- Displays the **Page Title**
- Provides context for the dashboard view

Purpose:

- Help users understand what the page represents
- Maintain consistency across different dashboard pages

KPI Cards Section

- Located at the top of the dashboard
- Displays key summary values

Purpose:

- Provide a quick overview of important indicators
- Allow users to grasp the overall situation instantly

Navigation & Slicer Panel

- Positioned on the left side
- Includes interactive filters (such as Years)

Purpose:

- Enable users to control and filter the data
- Apply selections across all visuals simultaneously

Main Visualization Area

The main dashboard area is divided into four visual sections:

Visual Function

Visual 1 Displays trends over time

Visual 2 Shows relationships between indicators

Visual 3 Provides supporting data insights Visual

4 Highlights a summary or key metric

Purpose:

- Organize information clearly
- Allow users to analyze data from different perspectives

2.4 User Experience (UX) Principles

The dashboard was designed based on the following UX principles:

Simplicity

- Only essential visuals are included
- Avoids clutter and unnecessary elements

Consistency

- Unified layout across all pages
- Consistent formatting of text and visuals

Readability

- Clear titles and labels
- Logical placement of elements

Interactivity

- Filters affect all visuals
- Users can explore different time periods easily

Visual Design Choices

Color Usage

- Consistent color scheme across the dashboard
- Colors used to improve clarity and differentiation between visuals

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Chart Types

Chart Type Purpose

Line Chart Display trends over time

Scatter Plot Show relationships

Chart Type Purpose

Cards Display key values

Area Chart Show comparisons over time

Gauge Represent summary values

Navigation Flow

The dashboard is designed to guide the user through a simple flow:

1. View summary information at the top
2. Use filters to select specific years
3. Analyze trends and comparisons
4. Draw conclusions from the visuals